

CSCE 633 MACHINE LEARNING · MODULE 10 · LECTURE NOTES

Probability and Uncertainty

Saying how sure the model is, and meaning it.

MODULE OUTCOMES

- Distinguish maximum likelihood from MAP estimation.
- Explain how a prior acts as regularisation.
- Explain naive Bayes and why it works despite its assumption.
- Distinguish aleatoric from epistemic uncertainty.
- Produce and interpret a prediction interval.

1 Likelihood, priors, and where losses come from

MLE: $\operatorname{argmax} P(D|\theta)$ **MAP:** $\operatorname{argmax} P(D|\theta)P(\theta)$ **Bayes:** $P(\theta|D) \propto P(D|\theta)P(\theta)$

Maximum likelihood chooses the parameters that make the observed data most probable. **MAP** weights that by a prior over parameters — and **taking the logarithm turns the prior into an additive penalty term**, which is where regularisation comes from. **The full Bayesian treatment** keeps the whole posterior distribution rather than a single point, which is expensive and is what genuine uncertainty estimates (§3) require.

YOUR LOSS FUNCTION WAS ALWAYS A LIKELIHOOD

Minimising squared error is maximum likelihood under additive Gaussian noise. Write down the Gaussian density, take its logarithm, discard the constants, and the squared term is what remains. **The choice of squared error was a distributional assumption**, made implicitly.

Minimising cross-entropy is maximum likelihood under a Bernoulli model — which is exactly how it was derived in Module 02 §3, where it appeared as 'the negative log-likelihood' without the implication being drawn out.

And the penalties of Module 04 are priors. A zero-mean Gaussian prior on the weights gives, through the log, a sum of squared weights — **ridge regression**. A Laplace prior, which is peaked at zero with heavy tails, gives a sum of absolute values — **lasso**, and **its peak at zero is the probabilistic counterpart of the geometric corner argument** of Module 04 §1.

So Module 04's regularisation was a probabilistic statement all along — 'I believe the weights are small' — and λ is how strongly you believe it, with large λ corresponding to a narrow prior. **This retroactively explains why regularisation helps rather than merely that it does**, and it is the main reason this module is placed where it is rather than first.

2 Naive Bayes

IT WORKS DESPITE BEING WRONG, AND THAT IS INSTRUCTIVE

Naive Bayes assumes every feature is conditionally independent of every other given the class. For text classification this asserts that word order and word co-occurrence carry no information about the class — which is plainly, obviously false.

Yet it classifies text well, and did so as the standard spam filter for years. The reason is that classification requires only the *argmax*: the estimated class probabilities can be badly wrong in magnitude while their ordering remains correct, and the ordering is all the decision uses.

So it is a good classifier and a terrible probability estimator. Because it multiplies many correlated pieces of evidence as though they were independent, it over-counts and its outputs are driven toward 0 and 1. Those numbers must never be used as probabilities (Module 05 §3) — and they frequently are, because they look like probabilities.

And it is extremely fast, needs very little data, handles very high dimensions, and trains in a single pass. Which is why it survived as the standard text baseline for a long time after better methods existed — and it is still the right first thing to try on a text problem, as a baseline in Module 01 §4's sense.

3 Two kinds of uncertainty

	Aleatoric	Epistemic
What it is	Noise inherent in the problem itself. Irreducible randomness in the data-generating process.	Ignorance arising from limited data. The model does not know because it has not been shown enough.
Example	The outcome of a fair coin flip. Measurement noise in a sensor.	A region of feature space the training set never covered — a demographic, a season, a failure mode.
Effect of more data	Does not reduce it at all.	Reduces it, which is why the distinction is actionable.
Where it appears in Module 03	The σ^2 term — the irreducible floor.	The bias and variance terms.
Correct response	Report it honestly; do not promise precision the problem does not permit.	Collect data in that region, or abstain from predicting there.

A model that cannot distinguish these cannot say 'I have never seen anything like this before' — which is frequently the most useful thing it could say, and is the basis of every sensible deferral or human-review policy.

A SOFTMAX OUTPUT IS NOT A CONFIDENCE

A classifier presented with an input unlike anything in its training data will still produce a confident-looking distribution over its classes. The softmax is constrained to sum to one, so it has no way to express 'none of these' — the probability mass has to go somewhere.

So a high softmax output means 'most like this class, among the classes I know about', which is a relative statement. It does not mean 'I am confident this input is that class', which is what it is universally read as.

Deep networks are routinely confidently wrong on out-of-distribution input (Module 05 §3) — assigning 99% confidence to noise, to adversarially perturbed images, and to entirely unrelated categories. This has caused real failures in deployed systems, and it is a direct consequence of the architecture rather than a bug.

Obtaining genuine epistemic uncertainty requires additional machinery: deep ensembles (train several and measure disagreement), Monte Carlo dropout (keep dropout active at inference and sample), conformal prediction (§4), or a properly Bayesian treatment. All of them cost, and none of them is the default, which is why the problem persists.

4 Prediction intervals

- **Quantile regression.** Fit models for the 5th and 95th percentiles directly, using the pinball loss. Simple, works with any gradient-based learner, and **captures aleatoric uncertainty that varies with x** — heteroscedasticity — which a single variance estimate cannot.
- **Bootstrap.** Refit the model on many resamples and take the spread of predictions at each point. **Captures epistemic uncertainty** — how much the fit would move with different data — and costs as many fits as resamples.
- **Ensembles.** The disagreement among independently trained models is a usable and cheap epistemic estimate, and deep ensembles remain one of the strongest practical methods despite being conceptually trivial.
- **Conformal prediction.** Gives a **distribution-free, finite-sample coverage guarantee** from a held-out calibration set, assuming only exchangeability — no assumption about the model or the data distribution at all. **It wraps any model**, including one you did not train, and it is **considerably underused** relative to how strong and how cheap the guarantee is.
- **And then measure the coverage.** A nominal 90% interval should contain the truth about 90% of the time on held-out data — check it, over the whole range and within slices.

AN UNCALIBRATED INTERVAL IS WORSE THAN NONE

An interval presented as 90% that in fact contains the truth 60% of the time is actively misleading — it invites decisions that would not have been made in the absence of any interval at all, which makes it worse than silence.

So measure the empirical coverage on held-out data, exactly as classification calibration is measured with a reliability diagram (Module 05 §3). The procedure is the same and so is the reasoning.

And report the measured coverage alongside the nominal level. '90% prediction intervals, empirically covering 88% on the test set, and 71% in the smallest subgroup' is an honest statement that a reader can act on. **Coverage within slices matters** — an interval that covers well on average and badly for a subpopulation is a fairness problem (Module 13).

This is the same discipline as every other 'claiming honestly' section in this program — CSCE 735 on speedups, CSCE 620 on robustness, CSCE 605 on miscompiles, CSCE 678 on availability. **State what you measured, not what you assumed.**

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Murphy — Probabilistic Machine Learning: An Introduction (free PDF)	Free in full. The MLE/MAP/Bayes material of §1 and the uncertainty treatment of §3, done properly.

Resource	What it covers for this module
Rennie et al. — Tackling the Poor Assumptions of Naive Bayes Text Classifiers (free)	Why §2's assumption survives being false , and what to fix when it does not.
Kendall & Gal — What Uncertainties Do We Need in Bayesian Deep Learning? (free)	The aleatoric/epistemic distinction of §3 , with methods for estimating each.
Angelopoulos & Bates — A Gentle Introduction to Conformal Prediction (free)	The §4 method , explained without measure theory. Short, and it should be far better known.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Derive squared error from the Gaussian likelihood** and cross-entropy from the Bernoulli, by hand.
- 2 **Derive ridge regression as MAP with a Gaussian prior** and relate λ to the prior variance.
- 3 Do the same for lasso and the Laplace prior.
- 4 Implement naive Bayes for text and compare its accuracy against logistic regression.
- 5 **Plot naive Bayes' predicted probabilities** and show they are pushed to the extremes.
- 6 **Construct in-distribution and out-of-distribution inputs** and compare the softmax outputs. Report the confidences.
- 7 Train a deep ensemble and use the disagreement as an epistemic estimate. Check it is high off-distribution.
- 8 Implement quantile regression with the pinball loss and plot the intervals.
- 9 **Implement split conformal prediction** and verify the coverage guarantee empirically.
- 10 **Measure coverage within subgroups** and report the worst one.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Distinguish MLE, MAP, and the Bayesian posterior.
- 2 Which likelihood gives squared error, and which gives cross-entropy?
- 3 Which priors give ridge and lasso?
- 4 Why does naive Bayes classify well despite a false assumption, and what is it bad at?
- 5 Distinguish aleatoric from epistemic uncertainty and say what each calls for.
- 6 Why is a softmax output not a confidence?

- 7 Name four ways to produce an interval, and say which gives a guarantee.
- 8 Why is an uncalibrated interval worse than none?